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Education

University of British Columbia 2021 – 2027 (expected)
PhD in Strategy and Business Economics
Committee: Guillermo Marshall (co-chair), Michael Sullivan (co-chair), Ekaterina Khmelnitskaya, Chunhua Wu

University of Wisconsin–Madison 2019 – 2021
MSc in Economics

Beijing Institute of Technology 2015 – 2019
BComm in Accounting
BA in Economics (dual)

Research Interests

Industrial Organization, Platform Economy, Digital Economy, Machine Learning

Working Papers

Who Benefits from Platform Competition? Evidence from Ride-Hailing (Job Market Paper)

Abstract Competition among digital platforms may enhance efficiency by curtailing platforms' market power or reduce efficiency by diminishing network externalities. Furthermore, because platforms set fees on both sides jointly, the different effects of competition on market power and on network externalities can translate into opposite welfare effects across sides. I study the welfare effects of platform competition with an equilibrium model estimated on data covering the universe of ride-hailing transactions in a major Chinese city. The model incorporates platforms' matching and pricing, drivers' platform adoption and labor supply, and riders' trip requests. I find that concentration can raise efficiency, by up to 3.8% relative to the status quo. However, the impact of concentration differs substantially across the sides of the market. Platforms earn more profit in a more concentrated market, while drivers lose surplus, because concentration not only increases platforms' monopsony power but also raises drivers' productivity, which lowers platforms' demand for labor. Whether riders gain or lose depends crucially on the level of concentration. They gain from mild concentration but lose from severe concentration. Holding market structure fixed, promoting multi-homing improves welfare and benefits all sides of the market.

Algorithmic Price Discrimination (with Yang Yu)

Abstract Modern pricing algorithms exploit high-dimensional consumer information, making price discrimination subtle and difficult for consumers and regulators to detect. We propose a machine-learning-based method to test whether a given piece of consumer information is used to price-discriminate. Built on a double/debiased machine-learning framework, the method removes bias from model misspecification and overfitting. We apply it to test whether a ride-hailing platform charges

frequent riders differently from occasional riders. A naive linear analysis produces persistent false positives, whereas our method finds no statistically significant difference in prices across riders.

Dual Role on Shoulders: Who Benefits from Platform Exclusivity?

Abstract A vertically integrated digital platform plays a dual role: it operates a marketplace for third-party products while selling its own first-party products on that marketplace. This paper examines whether such a platform should keep its first-party products exclusive to its own marketplace, and what this controversial practice implies for welfare. Using a model of Nash bargaining and Bertrand competition, I show that giving up exclusivity raises platform profits but lowers consumer surplus. The rationale is that selling first-party products on rival platforms serves as a collusive device: it softens competition for platform usage and ultimately leads to higher product prices.

Teaching Experience

Instructor

COMM 295 Managerial Economics — 2026 Summer

Teaching Assistant

COMM 491 Strategic Management in Business — 2023 Fall

Research and Professional Experience

DiDi — Beijing, China

Pricing Researcher — Oct 2024 – Dec 2025

Use machine learning to detect price discrimination and improve pricing fairness in pricing algorithms

Research Assistant

Prof. James Brander — 2022 – 2023

Prof. Ekaterina Khmelnitskaya — 2023 – 2024

UBC IO Workshop

Organizer — 2022 – 2024

Conferences and Summer Schools

2025: Luohan Academy Annual Digital Economy Conference

2024: Econometric Society Summer School in Dynamic Structural Econometrics (DSE)

2023: Competition & Regulation European Summer School and Conference (CRESSE)

Awards and Fellowships

President's Academic Excellence Initiative PhD Award — 2021 – 2026

Dean Earle D. MacPhee Memorial Fellowship in Commerce and Business Administration — 2021 – 2026

Sauder School of Business Graduate Award — 2021 – 2022

C.K. Choi Fellowship in Business Administration — 2021

Other Information

Programming: Python, Julia, SQL, R
Languages: English (Fluent), Mandarin (Native)

References

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